

Attention Is All You Need: Discovering Research Gaps in Neural Sequence Models

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks in an encoder-decoder configuration. In this paper, we propose the Transformer, an architecture eschewing recurrence and relying entirely on an attention mechanism to draw global dependencies between input and output. The Transformer allows for significantly more parallelization and establishes a new state of the art in machine translation quality.

1. Introduction

Recurrent neural networks, long short-term memory, and gated recurrent networks have been firmly established as state-of-the-art approaches in sequential language modeling and sequence transduction such as neural machine translation. Recurrent models typically factor computation along the symbol positions of the input and output sequences. Aligning positions to steps in computation time precludes parallelization within training examples, which becomes critical at longer sequence lengths.

Attention mechanisms have become an integral part of compelling sequence modeling and transduction models in various tasks, allowing modeling of dependencies without regard to their distance in the input or output sequences. In this work we propose the Transformer, a model architecture eschewing recurrence.

2. Methodology

The Transformer follows an overall encoder-decoder architecture using stacked self-attention and point-wise, fully connected layers for both the encoder and decoder. The encoder is composed of a stack of $N = 6$ identical layers. Each layer has two sub-layers: a multi-head self-attention mechanism and a position-wise feed-forward network. We employ residual connections around each of the sub-layers followed by layer normalization.

3. Limitations

While self-attention allows for full sequence parallelization, its computational complexity scales quadratically $O(N^2)$ with respect to input context length. This limitation presents a major bottleneck for ultra-long scientific document processing and extensive document-level context modeling. Furthermore, the lack of inherent inductive bias for temporal sequence ordering requires explicit positional encodings, which can degrade when evaluating on sequences longer than those encountered during training.

4. Conclusion

In this work, we presented the Transformer, the first sequence transduction model based entirely on attention. We plan to extend the Transformer to problems involving input and output modalities other than text, such as images and audio, and investigate localized attention mechanisms to address long sequences.

References

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